

Alex Taylor Wilk
 Alex Taylor Architecture
 241 Central Park West (13B)
 New York, NY 10024

08/15/2025

Re: Private Residence
 135 East 18th Street
 New York, NY 10003
 Preliminary MEP scope for rough budget

Ms. Wilk:

Below please find the preliminary mechanical, plumbing, electrical and sprinkler scope of work for preliminary budget numbers. Scope of work subject to change and full pricing must be provided based on fully developed engineered drawings.

Mechanical:

Heating & Cooling

(1) COND.
 REQUESTED
 3 PHASE
 4'X2.5' BY
 6FT TALL
 (WITHOUT
 DUNNAGE)

- (1) LG 10-ton heat recovery VRF system model ZRUM120BTE6
- ~~(2) LG 4 port heat recovery units, model PRHR043C~~
- ~~5~~ (9) Zones: Each with a fully ducted air handler mounted with secondary pans and water sensors. Air handlers to be a mix of Multi-Position (Vertical) units and Mid Static Ducted units.
- Provide ~~5~~ (9) LG deluxe wired wi-fi thermostats, model PREMTA201 and (9) LG remote temperature button sensors, model ZRTBS01.

Ducting & Outlets

INCOMING
 WOULD
 SING PHASE
 208 V
 400AMP 3
 PHASE
 VERIFY AMP
 IN SINGLE

- Provide 1" internally lined hard duct and plenum boxes throughout. Flex duct will not be permitted.
 - Supply - max 700 FPM; Return – max 600 FPM.
- Outlets to be Advanced Architectural Grille Model AAG-100, with plaster J-Bead frames.
 - Sized for a noise criteria no more than 20.

Exhaust & make up air

VS LARGER
 SING PHASE
 UNIT.

COME INTO

OR
 SING.
 PHASE
 POWER
 WITH TWO
 CONDENSE
 RS. EACH

- (6) Ceiling bathroom exhaust fans, Panasonic model FV-0511VK3
 - Provide Fittes flush exhaust mount [luxu] or custom 12x12 punched metal grille (by others) for all fans.
 - Fans to be ducted to roof with (1) bath exhaust roof cap.
- (1) 900 CFM Range exhaust hood by others to be ducted to the roof. Fan may be in hood, inline or roof mounted dependent on hood selection.

- (1) 600 CFM Range Makeup air system with heat, Electro-Industries model EM-MD10-240-1-10.
- Basement Dryer to be ventless
- 4th floor dryer to be vented to the roof.

Electric Radiant Floor

- Provide Nu-Heat radiant floor mats in all full bathrooms with signature Wi-Fi thermostats.

Plumbing

Incoming Services (dependent on inspection of material and condition)

- Existing 2½" sprinkler service to remain
 - New DEP approved DCDA
- Existing 1" domestic water service assumed to remain (material to be verified)
 - New DEP approved RPZA assembly with gravity drain
 - New NYC approved water meter
- Existing 1½" Gas service to remain
 - Provide new gas meter
- Existing combined storm / sanitary to city main to remain.
 - Line is to be scoped for existing conditions prior to work.
 - Provide New sanitary house trap and FAI.
 - Provide new storm piping within building with new storm trap to connect to existing line after new sanitary house trap.

Material

- Sanitary and vent lines to be domestic no-hub cast iron
- Domestic water to be copper, Type "L" throughout
 - Insulate all domestic water lines per 2020 NYC energy code.
- Gas piping to be steel pipe.

Domestic water

- (1) new gas domestic hot water tank, AO Smith model BTH-199.
 - Condensing direct vent model – Provide 4" ULC-S636 intake and vent pipes.
- Provide hot water recirculation line and pump.
- Provide allowance for domestic water boost pump, Grundfos model CMBE.
 - Existing pressure to be verified.

Sanitary

- Provide allowance for duplex sewage ejector system for Powder C06, Liberty Pumps model ProVore 680.
 - Existing house trap is below grade within mechanical vault – sanitary line elevation to be verified.

Sprinkler

Incoming service

- Existing 2½” incoming sprinkler service to remain (dependent on hydrant flow test results and hydraulic calculations)
 - Provide new 2½” house control valve
 - Provide new DCDA
 - Provide new Reliable commercial riser valve set model CR2-2.5”
 - Riser flow switch and valves to be monitored by central station security system or provide audible bell.

Sprinklers

- Ceiling – Flat concealed sprinkler heads, Reliable model RFC49
- Sidewall – Flat concealed sprinkler head, Reliable model RFS42
- Head quantities:
 - Cellar – 8
 - 1st Floor – 8
 - 2nd floor – 6
 - 3rd floor – 11
 - 4th floor – 10
 - Bulkhead – 2

Electrical

Provide new 400 amp 3-phase electrical service

- New trans “s” meter switch combination, Nav-Tech model TS/SW 400-240
 - 208/120v/3phase/4 wire – 400 amp rate / copper buss
- New Main distribution panel Square D I-line model HCN14524
- (3) power panels Square D model NQ454

Please feel free to contact me if you have any questions or comments with regards to the above responses.

Kind Regards,



Kyle Moody, PE
RJD Engineering LLC